Research Report | Commercial Profit Intelligence

Synthetic-data demonstration of customer, product, discount, geographic and operations analytics

1. Research objective

Develop and evaluate a reproducible interactive decision-support prototype for identifying line-level revenue, recorded profit, margin, discounts, customer concentration and regional variation. The project demonstrates a transparent analytical workflow rather than validating business performance at any named company.

2. Data provenance and reproducibility

The public dataset is generated entirely by `generate_demo_data.py` using a fixed random seed and contains 16,000 transaction lines. Customer IDs and location labels are synthetic. It contains no original company records. Reproduce the results by running the generator and then the analytics tests. No independent real-world accuracy claim follows from this constructed dataset.

3. Methods and formulas

For each line: gross sales = Sales; discount = Order Item Discount; net sales = Order Item Total; profit = Order Profit Per Order. A line is financially valid when the key fields are present, gross and net sales are positive, discount is nonnegative and |gross - discount - net| is at most 0.05 CU. Weighted net margin is 100 x SUM(profit) / SUM(net sales). Loss-making lines have profit below zero. Aggregations group the same valid filtered population.

4. Demonstration results

Metric	Result
Synthetic transaction lines	16,000
Gross sales	2,305,430.00 CU
Discounts	216,236.90 CU
Net revenue	2,089,193.10 CU
Recorded profit	276,686.22 CU
Weighted net margin	13.24%
Loss-making lines	3,595
Late-risk flag prevalence	48.01%

Category-level profitability (synthetic)

Category	Net revenue	Profit	Margin
Equipment	513,054	76,124	14.8%

Category	Net revenue	Profit	Margin
Apparel	438,118	58,288	13.3%
Electronics	395,986	55,675	14.1%
Accessories	380,974	44,083	11.6%
Footwear	361,061	42,516	11.8%

Market-level profitability (synthetic)

Market	Net revenue	Profit	Margin
LATAM	437,654	59,405	13.6%
USCA	419,471	56,410	13.4%
Europe	413,894	55,322	13.4%
Africa	409,608	54,190	13.2%
Pacific Asia	408,565	51,358	12.6%

5. Customer and operations analysis

Across the synthetic portfolio, 81 synthetic customer IDs have negative aggregate profit. Customer contribution tiers are descriptive rather than customer lifetime value estimates. Shipping mode is examined as an association with margin and the generated late-delivery risk flag; shipping costs are not independently observed as an audited field in the source schema.

Shipping mode	Profit (CU)	Late-risk flag
Standard Class	173,422	48.3%
Second Class	59,774	47.5%
First Class	38,455	48.1%
Same Day	5,036	47.1%

6. Discount sensitivity experiment

The scenario changes the line discount by a selected number of percentage points of gross sales, bounded to 0-100% of gross sales, while keeping quantities, unit economics and mix fixed. This is a mechanical accounting exercise, not a causal model or demand forecast.

Discount adjustment	Scenario profit (CU)	Profit change (CU)
-10 percentage points	417,298.82	140,612.60
-5 percentage points	359,334.58	82,648.36
+0 percentage points	276,686.22	0.00
+5 percentage points	161,414.72	-115,271.50
+10 percentage points	46,143.22	-230,543.00

7. Engineering and quality checks

The Streamlit application exposes seven linked sections with market, region, segment, category, product and shipping filters; CSV exports; scenario slider; and data-quality audit. Automated unit tests verify reconciliation, KPIs, aggregation, scenario behavior, and chart-data helpers. A dashboard smoke test executes the complete seven-tab application using a test stand-in for Streamlit; it does not replace a real browser test of the deployed service.

8. Limitations and ethical publication

All reported figures in this public paper are synthetic. The constructed dataset cannot validate operational conclusions, discount causation, temporal trends, actual customer retention or corporate performance. The original provided company dataset and original private analyses are not included in the public release. Any separate private deliverables must comply with the dataset provider's access and disclosure conditions.

9. Practical next steps

For a permitted real-data pilot: obtain approved data access; validate financial definitions and identifiers; incorporate dates and cost components; test segment stability; evaluate intervention impacts prospectively; and document data governance, deployment controls and reproducibility.